

 L12_S1_A.pdf Example Lecture***cture Only).pdf

Task: Generate a lecture scribe intended as exam-oriented reference material. The scribe must be faithful to the provided context and suitable for closed-notes exam revision.

Context rules: Use only the attached context (lecture slides and relevant textbook excerpts). Do not introduce any material, intuition, examples, or explanations that are not present in the provided context.

Single-response rule: Generate the complete lecture scribe in one response only. Do not ask questions, clarifications, confirmations, or request additional input. Do not provide options or commentary.

Structural constraint: The lecture scribe must strictly follow the structure shown in the attached "Example Lecture Scribe (Structure Only)" PDF. No reordering, additions, or deletions of sections are permitted.

The scribe must prioritize correctness over style; follow the lecture's structure and scope; include definitions, notation, assumptions, and stated results; include proofs, derivations, and worked examples only if they appear in the provided context; and maintain a formal, academic tone.



[cite_start]Lecture Title: Transformation of Random Variables and Functions of Two Random Variables [cite: 30, 34] [cite_start]Course: CSE400 - Fundamentals of Probability in Computing [cite: 2, 43] Purpose: Exam-oriented lecture scribe for revision

1. Overview

This lecture introduces the main concepts and results relevant to the transformation of random variables (RVs). [cite_start]The discussion covers techniques for finding the probability density function (PDF) of a new random variable Y derived from a known random variable X via a transformation $y = g(x)$ [cite: 21, 23, 24]. [cite_start]Furthermore, the lecture expands to functions of two random variables, specifically the derivation of distributions for $Z = g(X, Y)$, with a detailed focus on the sum of random variables ($Z = X + Y$)[cite: 34, 40].

2. Definitions and Notation

Definition 2.1 (PDF and CDF of Known RV): [cite_start]Let X be a random variable where the Probability Density Function (PDF), denoted as $f_X(x)$, and the Cumulative Distribution Function (CDF), denoted as $F_X(x)$, are known a priori[cite: 1, 8].

Definition 2.2 (Transformation of a Single RV): [cite_start]A new random variable Y is defined by the function $y = g(x)$, where $g(\cdot)$ is a transformation function[cite: 21, 28]. [cite_start]The objective is to find the PDF of Y , denoted as $f_Y(y)$ [cite: 25, 38].

Definition 2.3 (Function of Two RVs): [cite_start]A new random variable Z is defined by a function of two random variables X and Y , such that $Z = g(X, Y)$ [cite: 16]. Common examples include:

- [cite_start]Sum: $Z = X + Y$ [cite: 16].
- [cite_start]Difference: $Z = X - Y$ [cite: 211].
- [cite_start]Ratio: $Z = \frac{X}{Y}$ [cite: 223].

Definition 2.4 (Convolution): [cite_start]When $Z = X + Y$ and X, Y are independent, the PDF of Z is the convolution of the PDFs of X and Y [cite: 111, 154].

3. Assumptions and Conditions

The following assumptions are taken as given throughout the lecture:

- [cite_start]**Assumption A:** For the single variable transformation $y = g(x)$, the function is assumed to be monotonic (e.g., $\log(\cdot)$)[cite: 29]. [cite_start]This implies the existence of an inverse function $x = g^{-1}(y)$ [cite: 48].
- [cite_start]**Assumption B:** In specific derivations regarding the sum of random variables, X and Y may be assumed to be independent random variables[cite: 99, 102].
- [cite_start]**Assumption C:** The PDFs $f_X(x)$ and $f_Y(y)$ are known[cite: 205].

4. Main Result / Theorem

Theorem 4.1 (PDF of Transformed RV $Y = g(X)$): For a monotonic transformation $y = g(x)$, the PDF of Y is given by: $f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$ This is also expressed as

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)} \quad [cite_start] [cite: 52, 71, 75].$$

Theorem 4.2 (PDF of Sum of Two RVs $Z = X + Y$): The PDF of $Z = X + Y$ is determined by the integral: $f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx$ [cite_start][cite: 148]. If X and Y are independent, this simplifies to: $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x) dx$ [cite_start][cite: 154].

5. Proof/ Derivation

Proof Sketch for Theorem 4.1 (Single Variable Transformation):

- **Step 1:** Start with the definition of the CDF of Y : $F_Y(y) = Pr(Y \leq y) = Pr(g(X) \leq y)$ [cite_start][cite: 60].
- **Step 2:** Apply the inverse function. $Pr(g(X) \leq y) = Pr(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$ [cite_start][cite: 61].
- **Step 3:** Differentiate the CDF with respect to y to obtain the PDF: $f_Y(y) = \frac{d}{dy}[F_X(g^{-1}(y))]$
Using the chain rule: $f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y)$ [cite_start][cite: 52, 54].
[cite_start]Note: For decreasing functions, absolute value bars are added to ensure the PDF is non-negative[cite: 71, 90].

Proof Sketch for Theorem 4.2 (Sum of RVs):

- **Step 1:** Start with the CDF of Z : $F_Z(z) = Pr(Z \leq z) = Pr(X + Y \leq z)$ [cite_start][cite: 121].
- **Step 2:** Express this as a double integral over the region where $x + y \leq z$ (i.e., $y \leq z - x$):
 $F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{XY}(x, y) dy dx$ [cite_start][cite: 123, 124].
- **Step 3:** Differentiate $F_Z(z)$ with respect to z to find $f_Z(z)$. [cite_start]This application utilizes the Leibniz integral rule[cite: 131, 132]: $f_Z(z) = \frac{d}{dz} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{z-x} f_{XY}(x, y) dy \right) dx$ The result yields: $f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx$ [cite_start][cite: 143, 148].

6. Worked Example

Example 6.1 (Sine Transformation):

- **Problem:** Let X be a Uniformly distributed RV over $(-1, 1)$. [cite_start]Find the PDF of $Y = \sin\left(\frac{\pi X}{2}\right)$ [cite: 80].
- **Solution:**

1. Define $f_X(x)$: $f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$ [cite_start][cite: 82].

2. Invert the transformation: $y = \sin\left(\frac{\pi x}{2}\right) \implies x = \frac{2}{\pi} \sin^{-1}(y)$ [cite_start][cite: 83, 84].

3. Differentiate: $\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$ [cite_start][cite: 85].

4. Apply Theorem 4.1: $f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right|$ $f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}} = \frac{1}{\pi \sqrt{1-y^2}}$
[cite_start]for $-1 < y < 1$ [cite: 92, 93].

Example 6.2 (Sum of Independent Normal RVs):

- **Problem:** Let $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ be independent RVs. [cite_start]Prove that $Z = X + Y \sim N(0, 2)$ [cite: 102, 103].
- **Solution:**
 1. **Setup the Convolution Integral:** $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$ [cite_start][cite: 154].
 2. **Substitute PDFs:** $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, $f_Y(z - x) = \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2}$ [cite_start][cite: 160, 169].
 3. **Evaluate Integral:** $f_Z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 + (z-x)^2)} dx$ By completing the square in the exponent and integrating, the result simplifies to: $f_Z(z) = \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-\frac{z^2}{2(\sqrt{2})^2}}$ [cite_start][cite: 169, 174, 177].
 4. **Conclusion:** This matches the form of a Normal distribution $N(0, \sigma^2)$ where $\sigma^2 = 2$. [cite_start]Thus, $Z \sim N(0, 2)$ [cite: 177].

7. Summary

This lecture established the key definitions, assumptions, and results required for exam preparation regarding the transformation of random variables. [cite_start]It covered the method for finding the PDF of a single transformed variable $Y = g(X)$ via the CDF technique and differentiation[cite: 33, 52]. [cite_start]Additionally, it derived the convolution integral for the sum of two random variables $Z = X + Y$ and demonstrated these concepts through Uniform and Normal distribution examples[cite: 37, 103]. All material presented above is directly traceable to the lecture content.